

2. A 30 kW gas boiler is used for **700 hours** in 1 year.

(a) (i) Use the equation:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the units used in 1 year.

[2]

units used = kWh

(ii) Use the equation:

$$\text{cost} = \text{units used} \times \text{cost per unit}$$

to calculate the cost of using the boiler for 700 hours.

[2]

One unit of gas costs £0.12.

cost = £

(iii) Use your answer in (a)(ii) to calculate the cost of using the boiler for **1 hour**.

[1]

cost = £

(b) A solar water heating system costs £3600.

It is estimated to reduce gas consumption so that the boiler is used for only 650 hours instead of 700 hours.

(i) Calculate the saving made in 1 year by using the boiler for 50 hours less.

[1]

saving = £

(ii) Calculate the payback time of the cost of the solar water heating system.

[1]

payback time = years

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